

Name: _____ Date: _____

Period: _____

Spiral Review

1. Complete the square to rewrite $x^2 - 6x$ in the form $(x - h)^2 + c$. What is h , and what constant c must you add?

$(x - 3)^2 - 9$; $h = 3$; $c = -9$ (i.e., subtract 9 to balance)

2. A circle is centered at $(2, -3)$ with radius 5. Write its equation in standard form.

$(x - 2)^2 + (y + 3)^2 = 25$

3. Consider the equation $\frac{(x - 1)^2}{9} + \frac{(y + 2)^2}{4} = 1$.

(a) What is the center of this ellipse? $(1, -2)$

(b) What is the horizontal radius a ? $a = 3$ (since $a^2 = 9$)

(c) What is the vertical radius b ? $b = 2$ (since $b^2 = 4$)

Teacher Note

Common errors to watch: students may report $a = 9$ (forgetting to take the square root) or read the center as $(1, 2)$ (flipping the signs of h and k). Address both before moving into the lesson proper.